Shortest-side skeletons of edge-perimeter-minimizing polyhexes

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Abstract

A polyhex is a finite connected set of cells of the hexagonal tiling, and its edge perimeter is the number of unit edges separating it from its complement. The minimum edge perimeter for a given number of cells is classical, and so is the existence of one nested sequence of minimizers, one for each size. We prove a structural statement about *all* minimizers: deleting any shortest extreme line from a minimizer with at least three cells leaves a minimizer whose minimum edge semiperimeter (the *threshold* of its size) has dropped by exactly one, and the deleted line can be restored one cell at a time through configurations whose edge perimeter exceeds the minimum for their size by at most two. Consequently every minimizer lies on a chain of minimizers linked by such deletions, and every minimizer can be built from a single cell by adding one adjacent cell at a time with edge-perimeter excess never above two. The constant two cannot be lowered: the skeleton of minimizers exists for every minimizer, but not every minimizer lies on a nested sequence of minimizers at the level of single cells. The proof rests on the tight hull of a configuration, its exact defect identity, and a rigidity statement at the plateau starts whose threshold is 3 or 5 modulo 6.

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1 Introduction

1.1 The question

Cells of the hexagonal tiling are the objects; a *polyhex* is a finite connected set of them, and its *edge perimeter* $p(S)$ is the number of unit edges with exactly one side in S. Writing $p_{\min}(N)$ for the least edge perimeter of an N-cell polyhex and

$$\operatorname{exc}(S) := p(S) - p_{\min}(|S|)$$

for the *excess*, a *minimizer* is a polyhex with $\operatorname{exc} = 0$.

The static problem — what is $p_{\min}(N)$, and what do minimizers look like — is classical. The question here is dynamic. A minimizer is a finished object; can it be *assembled*, one cell at a time, without ever leaving the neighbourhood of optimality? Formally, given an N-cell minimizer S, is there a chain

$$S_1 \subset S_2 \subset \cdots \subset S_N = S, \qquad |S_k| = k,$$

of polyhexes in which each step adds a single adjacent cell and every S_k has small excess?

Two remarks fix the difficulty. First, one cannot ask for $\operatorname{exc}(S_k) = 0$ throughout: we exhibit a 13-cell minimizer from which no single cell can be removed to leave a 12-cell minimizer. Second, the natural greedy reading of the question is false: it is not true that every configuration of excess at most two admits a one-cell deletion keeping the excess at most two, and a 31-cell directionally convex example in Remark 8.1 has no such downward move. Thus the entire class $\{\operatorname{exc} \le 2\}$ cannot serve as the state space of a one-cell downward induction with a universally available move.

1.2 Results

The answer is that the growth has a *skeleton* made of minimizers. For a polyhex S, the six *extreme lines* are the outermost occupied lines in the three lattice directions, and a *shortest* extreme line is one carrying the fewest cells of S. For a minimizer the cells of S on each extreme line form a contiguous run; that is a theorem (Theorem 4.4(d)), and Definition 2.8 states the notion on the class of polyhexes for which it holds. Write $g(N) := \lceil \sqrt{12N-3}\,\rceil$, so that $p_{\min}(N) = 2g(N)$ by a classical result recalled in Section 4. We call $g(N)$ the *threshold* of N: it is the least edge semiperimeter an N-cell polyhex can have, and it is constant on runs of consecutive N that we call *plateaus*. The excess of a polyhex S is

$$\operatorname{exc}(S) = p(S) - p_{\min}(|S|) = 2\big(\sigma(S) - g(|S|)\big),$$

where $\sigma(S) = p(S)/2$ is its edge semiperimeter.

Theorem 1.1 (shortest-side skeleton)**.** *Let S be an edge-perimeter minimizer with $|S| \geq 3$ and let Λ be any shortest extreme line of S. Then $S \setminus \Lambda$ is an edge-perimeter minimizer with $g(|S \setminus \Lambda|) = g(|S|) - 1$, and the cells of $S \cap \Lambda$ can be restored one at a time through polyhexes of excess at most two.*

Iterating, every minimizer S admits a chain $S = T_0 \supset T_1 \supset \cdots \supset T_k$ of minimizers, $|T_k| \leq 2$, each obtained from the previous one by deleting a shortest extreme line. Interpolating each step and reading the result backwards gives the growth statement.

Theorem 1.2 (one-cell growth)**.** *Let S be an N-cell polyhex of minimum edge perimeter. Then there are polyhexes $S_1 \subset \cdots \subset S_N = S$ with $|S_k| = k$ such that*

(a) for each $k < N$, the polyhex S_{k+1} is obtained from S_k by adding one cell adjacent to S_k, and

(b) $\operatorname{exc}(S_k) \leq 2$ for every k.

Equivalently, writing $\mathcal{M}_N$ for the N-cell minimizers and $\mathcal{R}_N$ for the N-cell polyhexes reachable from a single cell by one-cell growth with excess never above two, $\mathcal{M}_N \subseteq \mathcal{R}_N$: restricting candidate generation to such growth excludes no minimizer.

Theorem 1.3 (optimality of the constant)**.** *The constant two in Theorem 1.2 cannot be lowered.*

Since p and $p_{\min}$ are both even, the excess is always an even non-negative integer, so the only smaller uniform bound would be zero; the 13-cell configuration mentioned above rules that out. Theorems 1.1 and 1.3 together locate the cost precisely: the skeleton of minimizers can always be kept optimal, and the excess 2 is spent only while a single extreme line is being rebuilt cell by cell.

1.3 How the proof works

Three ingredients carry the argument. First, every minimizer is *directionally convex*: along each of the three lattice directions its trace on every line is a single run. This follows from a comparison with the *tight hull* of the configuration, the lattice hexagon spanned by its extreme lines, whose cell count satisfies an exact *defect identity* in terms of its six side lengths. The same identity yields the capacity function $M(t) = \lfloor (t^2+3)/12 \rfloor$, the largest number of cells that fit within an edge-semiperimeter budget of t, and with it the formula for $p_{\min}$.

Second, the six side lengths of the hull add up to exactly three more than the number of occupied lines, so a shortest extreme line of a minimizer with threshold σ carries at most $\lfloor (\sigma+3)/6 \rfloor$ cells.

A residue computation modulo 6 shows that deleting that many cells from a minimizer lands on a minimizer one threshold step down — except possibly when the threshold is 3 or 5 modulo 6, the minimizer sits at the very start of its plateau, and the shortest line is as long as the bound allows.

Third, if in that exceptional situation the shortest trace attained the bound, the hull would be forced to be a regular hexagon, or a regular hexagon with one pair of opposite sides lengthened by one, and its boundary would then force the minimizer to fill the hull completely; but the plateau start has strictly fewer cells than that hexagon. This *rigidity* closes the last case. Restoring the deleted line one cell at a time never raises the perimeter above its starting value, while the threshold has already dropped by one, which is where the excess 2 comes from.

1.4 The role of computation

No theorem in this paper depends on finite enumeration, and neither does Theorem 1.3. Computation appears in Section 9 for two purposes: to determine, by complete enumeration, that 13 is the smallest size at which a minimizer has no all-minimizer growth chain — a question the analytic arguments do not answer; and to re-check the statements of the paper at small sizes with implementations independent of their proofs.

1.5 Relation to previous work

The formula $p_{\min}(N) = 2\lceil\sqrt{12N-3}\rceil$ is due to Harary and Harborth [8], who determined the extremal animals of the three Euclidean tilings; it corresponds, through $p = 6N - 2b$ with b the number of adjacent pairs of cells, to Harborth's earlier formula $\lfloor 3N - \sqrt{12N-3}\rfloor$ [9] for the maximum number of touching pairs among N non-overlapping unit discs in the plane. We give a self-contained derivation in Section 4, because the proof needs the hull machinery that produces it and not merely the value.

That machinery is itself close to known ground, and we are explicit about where. The capacity function $M(t)$ coincides, after the substitution $p = t + 3$, with the maximum-size function for the *site* perimeter determined by Fülep and Sieben [7, Proposition 5.10], and our hull defect identity is the image of their polyiamond identity [7, Lemma 3.3] under the triangular–hexagonal duality of their Section 5. Even the one exceptional value is the same on both sides. What Section 3 adds is not these identities but their formulation in edge-perimeter variables, which is what the rest of the paper consumes.

Barequet and Ben-Shachar [2] study minimum *site*-perimeter animals and show that inflation — adding to an animal its entire perimeter layer — induces a bijection between the minimizers of one size and those of a larger size, thereby proving the constant-isomer phenomenon observed in the chemistry of benzenoid hydrocarbons. A companion paper [3] gives algorithms for counting minimum-site-perimeter animals. That line of work differs from the present one in three respects: the functional is the site perimeter rather than the edge perimeter; inflation acts on a configuration as a whole and advances the size by the size of its perimeter, rather than by one cell; and the optimality requirement is imposed at the two sizes it relates, not at every intermediate size. We do not identify the two notions of perimeter anywhere, and no result about one is transferred to the other.

The closest previous work on *all* minimizers is that of Davoli, Piovano and Stefanelli [6], who study the edge-isoperimetric problem on the triangular lattice — the same problem as ours, see the identification below — with the declared aim of characterizing every minimizer. Four items of that paper bear on this one. Their equation (12) records the value $\theta_n = 2\lceil\sqrt{12n-3}\rceil$ of the minimum edge perimeter. Their Theorem 1.1 characterizes the connected minimizers as the con-

nected configurations attaining equality in an isoperimetric inequality between a lattice area and a lattice perimeter. Their Proposition 3.3 proves that every minimizer is 3-convex, which is our directional convexity; we state this correspondence at Theorem 4.4 and keep our own proof, which runs through the hull identities rather than through the rearrangement argument used there. Their Theorem 1.2 attaches to each minimizer M_n the largest regular lattice hexagon H_r contained in it and shows that $|M_n \setminus H_r| \le K_t n^{3/4} + o(n^{3/4})$, and their Theorem 1.4 shows that the constant K_t is optimal; along the way (their inequality (62)) the numbers of occupied lattice layers outside H_r on the six sides are bounded in total by their perimeter functional. That result describes how far a minimizer can deviate from a hexagon, in their metric; the present paper describes what happens when one extreme line is deleted. The two auxiliary hexagons point in opposite directions — theirs is the largest regular hexagon *inside* the configuration, ours (Section 3) is the tight lattice hexagon *around* it, not necessarily regular — and, more to the point, none of the results of [6] cited above states that the removal of a shortest extreme line preserves minimality, nor the exact lower bound on the number of remaining cells that this requires. That bound, $|S| - \rho \ge M(\sigma - 2) + 1$ in the notation of Section 6, is one integer stronger than what the capacity count gives in two residue classes of σ modulo 6, and closing that gap by the rigidity statement of Proposition 5.3 is the additional argument this paper supplies. We do not claim that Theorems 1.1–1.3 could not be extracted from the structure theory of [6] by further work; we claim that they are not stated there and that the present proof does not use it.

Vainsencher and Bruckstein [14] study isoperimetrically optimal polyforms on the regular tilings, the hexagonal one included, and observe that a spiral construction attains the optimal perimeter at every size without exhausting the optimal shapes. That concerns the shapes at a fixed size; the present paper concerns how the minimizers at consecutive thresholds are related to one another.

A nested sequence of minimizers is not in itself new. The cell-adjacency graph of the hexagonal tiling is the triangular lattice with its unit-distance edges, and in that graph the edge boundary of a set of cells is exactly its edge perimeter, so Theorem 1.2 concerns the edge-isoperimetric problem of a specific 6-regular Cayley graph on $\mathbb{Z}^2$. That problem is solved [8] (see also [9, 10]), and it is solved together with a *nested solution*: an ordering of the lattice all of whose prefixes are optimal [11, Theorem 7.2]. Harary and Harborth exhibited such an ordering as a spiral, and Malen, Roldán and Toalá-Enríquez [12] later extended the spiral construction to Euclidean and hyperbolic $\{p, q\}$-animals, recovering the three Euclidean minimum-perimeter formulas as a corollary.

What a nested solution provides is *one* sequence, passing through one minimizer at each size. The present results reverse the quantifier, at two granularities. Theorem 1.1 says that *every* minimizer lies on a nested chain of minimizers when the chain is allowed to advance by one extreme line at a time. Theorem 1.3 says that this fails when the chain must advance by one cell at a time: not every minimizer lies on a nested sequence of minimizers at that granularity, and Theorem 1.2 says that relaxing optimality by a single step of the edge perimeter is exactly what restores reachability for all of them.

Whether nested solutions exist at all is a question of independent interest. Bollobás and Leader [4] gave the first example of a graph without them, and Barber and Erde [1] asked whether every Cayley graph on $\mathbb{Z}^d$ has one; counterexamples are now known for $d = 1$ [5] and, in a strong form, for every $d \ge 2$ [13]. The lattice considered here is on the positive side of that divide, so the results below move the issue from the existence of a nested solution to the reachability of prescribed minimizers within nested optimal sequences, and show that the answer depends on the granularity of the steps.

To our knowledge, no previous result provides, for *every* edge-perimeter minimizer, a one-cell growth sequence by adjacent cells all of whose intermediate stages have edge-perimeter excess at most two, nor a chain of minimizers linking every minimizer to a single cell by deletions of whole

extreme lines.

To state plainly what is claimed as new. The paper has six layers: (1) the minimum perimeter and the capacity function; (2) the tight-hull identities; (3) directional convexity of minimizers; (4) the shortest-side skeleton theorem (Theorem 1.1); (5) bounded-excess one-cell reachability of every minimizer (Theorem 1.2); (6) optimality of the constant 2 (Theorem 1.3). Layers (1)–(3) are classical or close to known results in the senses made explicit above, and are included, with proofs, only so that the argument is self-contained in the variables it needs. The contribution of the paper is layers (4)–(6).

1.6 Organization

Sections 2–4 fix the lattice, the edge perimeter, directional convexity, the tight hull and its identities, and the capacity function. Section 5 bounds the shortest extreme line and proves the rigidity statement at exceptional plateau starts. Section 6 proves the skeleton theorem and derives Theorem 1.2. Section 7 proves that the constant two is optimal. Section 8 records what the argument does not give, and Section 9 the computational checks.

2 The hexagonal lattice, runs, and directional convexity

Sections 2–4 fix the lattice, the perimeter functional, the convexity notion, the tight hull and its identities, and the extremal capacity function. The proof of the main theorem in Sections 5–6 reads all of its geometry from here.

2.1 Lattice and polyhexes

Cells of the hexagonal tiling are indexed by axial coordinates $(q, r) \in \mathbb{Z}^2$. We call q, r and

$$s := q + r$$

the three *keys* of a cell; in words, the third key of a cell is the sum of its two axial coordinates. Two cells are *adjacent* when they differ by one of the six vectors

$$d_0 = (1, 0), \quad d_1 = (0, 1), \quad d_2 = (-1, 1), \quad d_3 = (-1, 0), \quad d_4 = (0, -1), \quad d_5 = (1, -1).$$

A *polyhex* is a finite nonempty set of cells that is connected with respect to adjacency. We write $|S|$ for the number of cells of S.

The following four facts are immediate from the definitions and are used throughout without further comment.

Fact 2.1. Two distinct cells are adjacent if and only if they agree in exactly one key and differ by exactly 1 in the other two. In words: an adjacency step keeps one of the three keys fixed and changes each of the other two by one unit (the step $d_0 = (1, 0)$, for instance, sends (q, r, s) to $(q + 1, r, s + 1)$, so the two changing keys need not move in opposite senses).

Proof. Each d_j changes (q, r, s) by $(1, 0, 1)$, $(0, 1, 1)$, $(-1, 1, 0)$, $(-1, 0, -1)$, $(0, -1, -1)$ or $(1, -1, 0)$. There are twelve triples $(\Delta q, \Delta r, \Delta s)$ with one zero entry and two entries equal to ± 1; exactly six of them satisfy $\Delta s = \Delta q + \Delta r$, and they are the six just listed. Conversely a difference of that shape is one of the d_j. □

Fact 2.2. A *line* is a set of cells on which one key is constant. Two lines belonging to different keys meet in exactly one cell; two distinct lines belonging to the same key are disjoint.

Fact 2.3. One adjacency step changes each of q, r and s by at most 1.

Fact 2.4. For a polyhex S and each key, the set of values taken by that key on S is an interval of consecutive integers. In words: no key value inside the range of S can be skipped.

Proof. Let k be a key and let $x, y \in S$ realize two values of k. A path of adjacent cells inside S joins them, and by Fact 2.3 the value of k changes by at most one per step, so every intermediate integer occurs on the path. □

2.2 Edge perimeter, runs and directional convexity

Let $a(S)$ denote the number of unordered pairs of adjacent cells of S. The *edge perimeter* of S is

$$p(S) = 6|S| - 2a(S), \tag{1}$$

in words: six times the number of cells, minus twice the number of adjacent pairs. Equivalently, $p(S)$ counts the cell edges of S that are not shared with another cell of S; we call these the *boundary edges*. The *edge semiperimeter* is

$$\sigma(K) := \tfrac{1}{2}p(K)$$

for any finite set of cells K; in words, half the edge perimeter. That this is an integer is part of Lemma 2.6 below.

A line Λ is *occupied* by S when $S \cap \Lambda \neq \emptyset$; we call $S \cap \Lambda$ the *trace* of S on Λ. Cells of a line are linearly ordered (for a line of constant q by the value of r, for one of constant r by the value of q, for one of constant s by the value of q), and consecutive cells in that order are adjacent. A *run* of S on Λ is a maximal block of consecutive cells of the trace. Let

$$R(S) = \text{the total number of runs of } S \text{ over all lines}, \qquad \lambda(S) = \text{the number of occupied lines}.$$

Definition 2.5. A polyhex S is *directionally convex* when the trace of S on every occupied line is a single run. In words: along each of the three directions, S never leaves a line and comes back to it.

This is a lattice notion and not Euclidean convexity; it is the notion called *layer convexity* in parts of the literature on lattice animals.

Lemma 2.6. *For every finite set of cells S,*

$$p(S) = 2R(S) \;\geq\; 2\lambda(S), \tag{2}$$

in words: the edge perimeter equals twice the total number of runs, and the total number of runs is at least the number of occupied lines. In particular $p(S)$ is even and $\sigma(S) = R(S)$. For a polyhex, equality $R(S) = \lambda(S)$ holds if and only if S is directionally convex.

Proof. Let $e = \{x, y\}$ be a boundary edge, with $x \in S$ and $y \notin S$ the two cells it separates. By Fact 2.1 the cells x and y agree in exactly one key, hence they are consecutive cells of exactly one line Λ, and x is an end of the run of S on Λ containing it. Conversely a run has exactly two ends, and the cell immediately beyond each end lies outside S and determines a boundary edge; for a run consisting of a single cell the two ends are the two opposite sides of that cell, so they give two distinct boundary edges. This is a bijection between boundary edges and ends of runs, whence $p(S) = 2R(S)$.

Every occupied line carries at least one run, so $R(S) \geq \lambda(S)$, with equality exactly when every occupied line carries exactly one run, which is Definition 2.5. □

Remark 2.7 (edge perimeter versus site perimeter). The *site perimeter* of S is the number of cells outside S adjacent to at least one cell of S. It is a different functional from (1), and the two have different minimizers in general; results about one do not transfer to the other. Nothing in this paper depends on the site perimeter. We mention it only because part of the literature we cite in Section 3.2 is written in those terms.

2.3 Extreme support lines and traces

Let S be a polyhex. For each key, Fact 2.4 makes the occupied values an interval, so there is a smallest and a largest occupied line in each of the three families. These six lines are the *extreme lines* of S; we list them in the cyclic order

$$\Lambda_1 :\ q = q_{\min}, \quad \Lambda_2 :\ r = r_{\max}, \quad \Lambda_3 :\ s = s_{\max}, \quad \Lambda_4 :\ q = q_{\max}, \quad \Lambda_5 :\ r = r_{\min}, \quad \Lambda_6 :\ s = s_{\min},$$

so that Λ_i and Λ_{i+3} are the two extreme lines of one and the same family, indices being read in $\mathbb{Z}/6$. If a family occupies a single line then the two corresponding extreme lines coincide; all statements below are read with that degeneracy allowed.

Definition 2.8. Let S be a *directionally convex* polyhex. Its six *support traces* are the traces $S \cap \Lambda_i$, each a single run by Definition 2.5, and we write

$$r_i := |S \cap \Lambda_i| \ \geq 1, \qquad i \in \mathbb{Z}/6, \qquad \text{and} \qquad \rho(S) := \min_i r_i.$$

The six extreme lines of S will turn out to be the six side lines of its tight hull (Lemma 3.4(ii)); from then on the same symbol Λ_i serves both roles, and the start of Section 5 fixes the terminology for the two finite sets, side and support trace, that live on it.

Remark 2.9. Definition 2.8 is stated for directionally convex S only. On that domain $\rho(S)$ is at once the length of the shortest support trace and the length of the shortest extreme-line run. For a general polyhex a trace need not be a run, and we do not extend the definition.

3 Tight hulls and exact identities

3.1 Tight lattice hulls

Definition 3.1. Let $L \geq 1$ and $u, v, w \geq 0$ be integers satisfying the *admissibility* conditions

$$u + v \leq L - 1, \qquad v + w \leq L - 1, \qquad u + w \leq L - 1, \tag{3}$$

and put $A := L - u$, $B := L - v$, $C := L - w$, so that $A, B, C \geq 1$. The corresponding *lattice hexagon* is

$$H(L; u, v, w) := \{\, (q, r) :\ 0 \leq q \leq A - 1,\ \ 0 \leq r \leq B - 1,\ \ w \leq q + r \leq L - 1 \,\}.$$

In words: the triangle of side L with three corner triangles of sides u, v and w cut off. Degenerate shapes, in which one or more of the cut corners is empty or one of the three widths equals 1, are allowed.

Definition 3.2. The *tight hull* $H(S)$ of a polyhex S is the lattice hexagon determined by the extreme values of the three keys on S. Concretely, normalize by a translation so that $q_{\min} = r_{\min} = 0$ and set

$$A = 1 + q_{\max}, \quad B = 1 + r_{\max}, \quad L = 1 + s_{\max}, \quad w = s_{\min}, \quad u = L - A, \quad v = L - B.$$

For this to define a lattice hexagon in the sense of Definition 3.1, the four parameters must be admissible. They are.

Lemma 3.3 (the tight hull is admissible)**.** *For every polyhex S, the parameters L, u, v, w of Definition 3.2 satisfy $L \geq 1$, $u, v, w \geq 0$ and the three inequalities* (3)*. Hence $H(S) = H(L; u, v, w)$ is a lattice hexagon.*

Proof. After the normalization $q_{\min} = r_{\min} = 0$, every cell of S has $q, r \geq 0$, hence $s = q + r \geq 0$; so $w = s_{\min} \geq 0$ and $L = 1 + s_{\max} \geq 1$. In the notation of the definition, $u = s_{\max} - q_{\max}$ and $v = s_{\max} - r_{\max}$, and the four remaining conditions are the following four inequalities between extreme values, each read off from one well-chosen cell of S.

- $u \geq 0$ and $v \geq 0$: a cell with $q = q_{\max}$ has $s = q_{\max} + r \geq q_{\max}$, so $s_{\max} \geq q_{\max}$; likewise $s_{\max} \geq r_{\max}$.
- $u + v \leq L - 1$, that is $s_{\max} \leq q_{\max} + r_{\max}$: a cell with $s = s_{\max}$ has $s_{\max} = q + r \leq q_{\max} + r_{\max}$.
- $v + w \leq L - 1$, that is $s_{\min} \leq r_{\max}$: a cell with $q = q_{\min} = 0$ has $s = r \leq r_{\max}$, so $s_{\min} \leq r_{\max}$.
- $u + w \leq L - 1$, that is $s_{\min} \leq q_{\max}$: a cell with $r = r_{\min} = 0$ has $s = q \leq q_{\max}$.

Note that only the six extreme values are used, not connectedness. □

For a lattice hexagon H we write $\Lambda_1, \ldots, \Lambda_6$ for its six extreme lines in the cyclic order of Section 2.3, namely

$$q = 0, \quad r = B - 1, \quad s = L - 1, \quad q = A - 1, \quad r = 0, \quad s = w,$$

and call them the *side lines*; we write

$$\ell_i := |H \cap \Lambda_i|$$

for the six *side lengths*, $h := \min_i \ell_i$ for the shortest side length, and $\kappa_i := H \cap \Lambda_i \cap \Lambda_{i+1}$ for the *corner* cells. Finally we set

$$\delta(H) := \ell_1 - \ell_4, \qquad Q(H) := (\ell_1 - \ell_3)^2 + (\ell_3 - \ell_5)^2 + (\ell_5 - \ell_1)^2.$$

Lemma 3.4. *Let S be a polyhex and $H = H(S)$. Then*

(i) $S \subseteq H$;

(ii) S and H occupy exactly the same lines, so $\lambda(S) = \lambda(H)$;

(iii) each of the six side lines of H meets S.

Proof. (i) is immediate from Definition 3.2 and Lemma 3.3, since every cell of S satisfies the six inequalities defining H. For (ii), the lines occupied by S in the q-family are, by Fact 2.4, exactly $q = 0, \ldots, A - 1$, and likewise $r = 0, \ldots, B - 1$ and $s = w, \ldots, L - 1$; these are precisely the lines occupied by H, as read off from Definition 3.1. Assertion (iii) is the case of (ii) for the six extreme values: each of them is attained by some cell of S. □

Lemma 3.4 is the only place where connectedness of S is used in this section, and it is used through Fact 2.4: for a disconnected set the occupied key values need not form an interval, and (ii) fails.

3.2 Hull identities

Lemma 3.5. *Let $H = H(L; u, v, w)$ be a lattice hexagon. Then:*

(i) the six side lengths are

$$\ell_1 = L - v - w, \quad \ell_2 = v + 1, \quad \ell_3 = L - u - v, \quad \ell_4 = u + 1, \quad \ell_5 = L - u - w, \quad \ell_6 = w + 1,$$

and each side line meets H in a single run whose two end cells are the corners κ_{i-1} and κ_i;

(ii) H is a directionally convex polyhex with

$$\lambda(H) = \sigma(H) = A + B + C = 3L - (u + v + w);$$

(iii) the six side lengths satisfy

$$\sum_{i=1}^{6} \ell_i = \lambda(H) + 3, \tag{4}$$

in words: the six side lengths of the hull add up to three more than the number of lines the hull occupies; and the three differences across opposite sides agree,

$$\ell_1 - \ell_4 = \ell_3 - \ell_6 = \ell_5 - \ell_2 = \delta(H) = L - u - v - w - 1; \tag{5}$$

(iv) with $T(k) := k(k+1)/2$,

$$|H| = T(L) - T(u) - T(v) - T(w);$$

(v) (hull defect identity)

$$\lambda(H)^2 + 3 - 12|H| \;=\; 3\,\delta(H)^2 + 2\,Q(H) \;\geq\; 0, \tag{6}$$

in words: the square of the number of occupied lines, plus three, minus twelve times the number of cells, equals three times the square of the common opposite-side difference plus twice the imbalance $Q(H)$.

Proof. (i) On the line $q = 0$ the defining inequalities reduce to $w \leq r \leq \min(B-1, L-1) = B-1$, which is a nonempty interval because $w \leq B-1$ is the condition $v + w \leq L-1$ of (3); its length is $B - w = L - v - w$. On the line $r = B-1$ they reduce to $\max(0, w-B+1) \leq q \leq \min(A-1, v)$, and (3) gives $w - B + 1 \leq 0$ and $v \leq A-1$, so the interval is $0 \leq q \leq v$, of length $v+1$. The remaining four sides are obtained in the same way, each time using exactly one of the three inequalities of (3) at each end. Each side is an interval of consecutive cells of its line, hence a run; its two end cells are the two cells where a second defining inequality becomes an equality, that is, the two corners.

(ii) The occupied lines are $q = 0, \ldots, A-1$, $r = 0, \ldots, B-1$ and $s = w, \ldots, L-1$, which number $A + B + C$; on each of them the trace is an interval of consecutive cells by the same computation as in (i), so H is directionally convex and Lemma 2.6 gives $\sigma(H) = R(H) = \lambda(H)$. Connectedness follows since each trace is a run and consecutive traces meet a common line.

(iii) Summing the six expressions of (i),

$$\sum_i \ell_i = 3L - (u + v + w) + 3 = \lambda(H) + 3.$$

Using the side-length formulas of (i), the three opposite-side differences $\ell_1 - \ell_4$, $\ell_3 - \ell_6$, $\ell_5 - \ell_2$ of (5) are $(L - v - w) - (u + 1)$, $(L - u - v) - (w + 1)$ and $(L - u - w) - (v + 1)$, all equal to $L - u - v - w - 1$.

(iv) The triangle $\{q, r \geq 0,\ q+r \leq L-1\}$ has $T(L)$ cells. The three sets removed in Definition 3.1 are the corner triangles $q \geq A$, $r \geq B$ and $q + r \leq w - 1$, of $T(u)$, $T(v)$ and $T(w)$ cells respectively, and (3) makes them pairwise disjoint: a cell with $q \geq A$ and $r \geq B$ would have $q + r \geq A + B = 2L - u - v \geq L + 1$, which is impossible, and a cell with $q \geq A$ and $q + r \leq w - 1$ would give $r \leq w - 1 - A < 0$, and symmetrically.

(v) Put $U := u + v + w$ and $P := u^2 + v^2 + w^2$. By (ii) and (iv),

$$\lambda(H)^2 + 3 - 12|H| = (3L - U)^2 + 3 - 6L(L + 1) + 6P + 6U = 3L^2 - 6LU - 6L + U^2 + 6P + 6U + 3.$$

On the other side, $\delta(H) = L - U - 1$ and, by (i), $Q(H) = (u - w)^2 + (w - v)^2 + (v - u)^2 = 3P - U^2$, so

$$3\delta(H)^2 + 2Q(H) = 3(L - U - 1)^2 + 2(3P - U^2)$$

expands to the same expression. In words: both sides equal three times L^2, minus six times LU, minus six times L, plus U^2, plus six times P, plus six times U, plus three. Nonnegativity is clear from the right-hand side. □

Write $h := \min_i \ell_i$ for the shortest side length and

$$x_i := \ell_i - h \ \geq 0$$

for the *side excess*, the amount by which a hull side exceeds the shortest one.

Lemma 3.6 (opposite-side excess)**.** *If $\sum_i x_i \leq 2$ then $x_i = x_{i+3}$ for every i; in particular $\sum_i x_i$ is even, so $\sum_i x_i = 1$ is impossible, and if $\sum_i x_i = 2$ then exactly one pair of opposite sides has length $h + 1$ and the other four sides have length h.*

Proof. Since the x_i differ from the ℓ_i by the constant h, (5) gives $x_1 - x_4 = x_3 - x_6 = x_5 - x_2 = \delta(H)$. If $\delta(H) \geq 1$ then $x_1, x_3, x_5 \geq 1$ and $\sum x_i \geq 3$; if $\delta(H) \leq -1$ the same holds for x_2, x_4, x_6. Both contradict the hypothesis, so $\delta(H) = 0$ and $\sum x_i = 2(x_1 + x_3 + x_5)$. □

Lemma 3.7 (a complete boundary forces equality with the hull)**.** *Let S be a directionally convex polyhex and $H = H(S)$. If $H \cap \Lambda_i \subseteq S$ for all six i, then $S = H$.*

Proof. Fix a value q_0. By Definition 3.1 the cells of H on the line $q = q_0$ are those with r in the interval $\big[\max(0, w - q_0),\ \min(B - 1, L - 1 - q_0)\big]$. Its lower end has $r = 0$ or $q + r = w$, so it lies on Λ_5 or Λ_6; its upper end has $r = B - 1$ or $q + r = L - 1$, so it lies on Λ_2 or Λ_3. By hypothesis both ends lie in S, and the trace of S on this line is a run, so it contains everything between them. As q_0 was arbitrary, $H \subseteq S$; together with $S \subseteq H$ from Lemma 3.4(i), this gives $S = H$. □

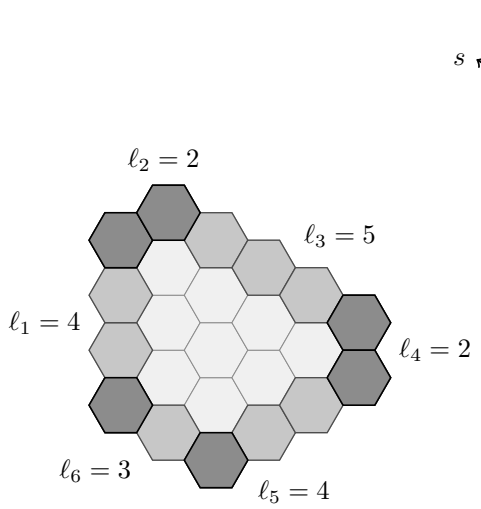


Figure 1: The tight hull $H(7; 1, 1, 2)$, with its six side lines shaded and the six corner cells $\kappa_1, \dots, \kappa_6$ darker. The side lengths are $\ell_1, \dots, \ell_6$ in the cyclic order of Section 2.3, namely $q_{\min}$, $r_{\max}$, $s_{\max}$, $q_{\max}$, $r_{\min}$, $s_{\min}$. Here $\sum_i \ell_i = 20 = \lambda(H) + 3$ and the three opposite differences all equal $\delta(H) = 2$; the shortest side length is $h = 2$.

Remark 3.8 (attribution). An equivalent defect identity is obtained from Fülep and Sieben's polyiamond identity [7, Lemma 3.3] through the triangular–hexagonal duality developed in their Section 5. In the same site-perimeter variables, [7, Proposition 5.10] yields exactly the capacity function $M(t)$ of Section 4 after the substitution $p = t+3$; the value $p = 7$ excluded there corresponds to the value $t = 4$ excluded in Remark 4.5 below. We nevertheless give the derivation here in edge-perimeter variables to keep the argument self-contained, and we do not identify the two settings: [7] is about the site perimeter (Remark 2.7).

4 Capacity and minimum edge perimeter

4.1 Peeling and extremal capacity

We first record the two operations on directionally convex polyhexes that the capacity computation needs: removing a whole support trace, and putting it back one cell at a time.

Lemma 4.1 (peeling)**.** *Let K be a directionally convex polyhex, let Λ be one of its six extreme lines, and let $K' := K \setminus \Lambda$. If $K' \neq \emptyset$, then*

(i) K' is a directionally convex polyhex with $\sigma(K') \le \sigma(K) - 1$;

(ii) the cells of $K \cap \Lambda$ can be ordered $c_1, \dots, c_k$ so that for every j the set $K' \cup \{c_1, \dots, c_j\}$ is a directionally convex polyhex of edge semiperimeter at most $\sigma(K)$.

Proof. Say Λ is the line $q = q_{\min}$; the other five cases are the same after permuting the roles of the keys and of the two senses.

(i) For each occupied value k of q let K_k be the trace of K on $q = k$; it is a run. If K is connected, then for consecutive values $k, k+1$ a path inside K joining a cell of K_k to a cell of K_{k+1} must, by Fact 2.3, contain a step from a cell with $q = k$ to a cell with $q = k+1$, so K_k and K_{k+1} are adjacent. Hence K', the union of the K_k with $k \ge q_{\min} + 1$, is connected. Its traces in the q-family

are traces of K, hence runs. Now fix a line of the r-family, $r = k$; its cells are (q, k), so the trace of K on it is a run if and only if its q-values form an interval, and by Fact 2.2 it meets Λ in at most the one cell $(q_{\min}, k)$. The trace of K' on this line is the trace of K with that cell, if present, removed. Since $q_{\min}$ is the least q-value occurring anywhere in K, it is the least q-value of the interval, so removing it leaves an interval, possibly empty. A line of the s-family, $s = k$, has cells $(q, k - q)$, again parametrized by q, and the same sentence applies verbatim. So all traces of K' are runs and K' is directionally convex. Finally Λ is occupied by K and not by K', so $\lambda(K') \le \lambda(K) - 1$, and by Lemma 2.6 $\sigma(K') = \lambda(K') \le \lambda(K) - 1 = \sigma(K) - 1$.

(ii) Since K is connected there is a cell $c_1 \in K \cap \Lambda$ adjacent to K'. Order the remaining cells of the run $K \cap \Lambda$ so that every initial segment $\{c_1, \dots, c_j\}$ is a block of consecutive cells of that run containing c_1. Write $K_j := K' \cup \{c_1, \dots, c_j\}$. Then K_j is connected, because c_1 is adjacent to K' and each later c_j is adjacent to a cell already inserted. Its trace on Λ is a block, hence a run; its trace on any other line of the q-family is a trace of K', hence a run. On a line of the r- or s-family, parametrized by q as in (i), the trace of K_j is the trace of K' there together with at most one inserted cell, which lies on Λ and so has $q = q_{\min}$. The trace of K on that line is a run containing this cell and, by (i), the trace of K' is that run with its least element $q_{\min}$ removed; putting the cell back restores the run of K. Hence every trace of K_j is a run, so K_j is directionally convex, and every line it occupies is occupied by K, whence $\sigma(K_j) = \lambda(K_j) \le \lambda(K) = \sigma(K)$. □

Corollary 4.2. *Let K be a directionally convex polyhex. For every j with $1 \le j \le |K|$ there is a directionally convex polyhex $K_j \subseteq K$ with $|K_j| = j$ and $\sigma(K_j) \le \sigma(K)$.*

Proof. Strong induction on $|K|$. For $|K| = 1$ there is nothing to prove. Let $|K| \ge 2$. Some extreme line Λ does not contain all of K: otherwise K would be contained in two extreme lines belonging to different families, which by Fact 2.2 meet in a single cell, contradicting $|K| \ge 2$. Apply Lemma 4.1 to that Λ. The sets K_j of part (ii) supply all sizes between $|K'|$ and $|K|$; the sizes below $|K'|$ are supplied by the inductive hypothesis applied to K', whose edge semiperimeter is at most $\sigma(K)$. □

Definition 4.3. For an integer $t \ge 0$ put

$$M(t) := \left\lfloor \frac{t^2 + 3}{12} \right\rfloor .$$

Thus $M(0) = M(1) = M(2) = 0$ and $M(3) = 1$. The function M is nondecreasing.

Theorem 4.4. *Let $N \ge 1$.*

(a) Every polyhex S satisfies $|S| \le M(\sigma(S))$: a configuration of edge semiperimeter t cannot have more than $M(t)$ cells.

(b) For every $t \ge 3$ there is a directionally convex polyhex K with $|K| = M(t)$ and $\sigma(K) \le t$.

(c) The minimum edge perimeter of an N-cell polyhex is

$$p_{\min}(N) = 2g(N), \qquad g(N) = \left\lceil \sqrt{12N - 3} \right\rceil .$$

(d) Every N-cell minimizer S is directionally convex and satisfies $\sigma(S) = \lambda(S) = g(N)$.

The directional convexity of minimizers in (d) is a known result. Davoli, Piovano and Stefanelli [6, Definition 3.2 and Proposition 3.3] prove that every minimizer of the edge-isoperimetric problem on the triangular lattice is 3-*convex*: for any two of its points on a common lattice line, all lattice

points between them belong to it. Under the identification of Section 1.5 this is exactly directional convexity. Their proof is a rearrangement argument, reordering the numbers of points on the parallel lines of one family; the proof below instead obtains (d) from the hull identities of Section 3 and the capacity bound (a), together with the additional information $\lambda(S) = g(N)$ that the rest of the paper consumes. We give it in full so that the paper is self-contained, not because a new proof of the convexity itself is needed.

Proof. (a) Let $H = H(S)$. By Lemma 3.4 and Lemma 3.5(v),

$$12|S| \le 12|H| = \lambda(H)^2 + 3 - \big(3\delta(H)^2 + 2Q(H)\big) \le \lambda(S)^2 + 3 \le \sigma(S)^2 + 3,$$

the last step by Lemma 2.6. Since $|S|$ is an integer, $|S| \le M(\sigma(S))$.

(b) For $t \equiv 3 \pmod 6$, say $t = 6m + 3$, take $K = H(3m + 1; m, m, m)$; for $t = 6m$ take $H(3m - 1; m - 1, m - 1, m - 1)$; for $t = 6m + 1$ take $H(3m; m, m, m - 1)$; for $t = 6m + 2$ take $H(3m + 1; m, m, m + 1)$; for $t = 6m + 4$ take $H(3m + 1; m, m, m - 1)$; for $t = 6m + 5$ take $H(3m + 2; m, m, m + 1)$. Here $m \ge 0$ for $t \equiv 3, 5 \pmod 6$ and $m \ge 1$ for the other four residues, which covers every $t \ge 3$ except $t = 4$; in each case (3) holds, Lemma 3.5(ii) gives $\sigma(K) = t$, and Lemma 3.5(iv),(v) give $12|K| = t^2 + 3 - D$ with $D \in \{0, 3, 4, 7\}$ according to the residue of t, so $|K| = M(t)$. The single case not covered by an admissible triple is $t = 4$; there $M(4) = 1$ and a single cell has edge semiperimeter $3 \le 4$.

(c) Put $t_N := \min\{t \ge 3 : \ M(t) \ge N\}$. By (a) every N-cell polyhex has $M(\sigma(S)) \ge N$, hence $\sigma(S) \ge t_N$. Conversely (b) produces a directionally convex K with $|K| = M(t_N) \ge N$ and $\sigma(K) \le t_N$, and Corollary 4.2 extracts from it an N-cell polyhex of edge semiperimeter at most t_N. Therefore $p_{\min}(N) = 2t_N$. Finally, for integers t,

$$M(t) \ge N \iff \frac{t^2 + 3}{12} \ge N \iff t \ge \sqrt{12N - 3},$$

and $\sqrt{12N - 3} \ge 3$ for $N \ge 1$, so $t_N = g(N)$.

(d) By (c), $\sigma(S) = g(N)$. Let $H = H(S)$. Applying (a) to H, which is directionally convex with $\sigma(H) = \lambda(H)$, gives $M(\lambda(H)) \ge |H| \ge N$ and therefore $\lambda(H) \ge t_N = g(N)$. Combining with Lemmas 2.6 and 3.4,

$$g(N) = \sigma(S) = R(S) \ \ge \ \lambda(S) = \lambda(H) \ \ge \ g(N),$$

so $R(S) = \lambda(S)$ and S is directionally convex. □

Remark 4.5. Parts (a) and (b) together say exactly that

$$M(t) = \max\{\, |K| : \ K \text{ a polyhex with } \sigma(K) \le t \,\},$$

in words: $M(t)$ is the largest number of cells that fit within an edge-semiperimeter budget of t. The budget form is the one the rest of the paper uses, and the inequality in (b) is not a weakening: for every $t \ge 3$ other than $t = 4$ the hexagons listed in the proof do satisfy $\sigma(K) = t$, and the value $t = 4$ is not the edge semiperimeter of any polyhex at all. Indeed a polyhex K with $\sigma(K) = 4$ would have $|K| \le M(4) = 1$ by (a), while the one-cell polyhex has edge semiperimeter 3.

4.2 The plateau bridge

The function $N \mapsto g(N)$ is nondecreasing and takes each of its values on a block of consecutive integers. For γ in the range of g let

$$N_b(\gamma) := \min\{\, N \geq 1 :\ g(N) = \gamma \,\}$$

be the first cell count at which the minimum edge semiperimeter equals γ.

Lemma 4.6. *For every γ in the range of g,*

$$N_b(\gamma) = M(\gamma - 1) + 1, \tag{7}$$

in words: the plateau on which the minimum edge semiperimeter equals γ begins one cell after the capacity of edge semiperimeter $\gamma - 1$ has been exhausted.

Proof. For integers $N \geq 1$ and $t \geq 3$ we have, as in the proof of Theorem 4.4(c), $g(N) \leq t$ if and only if $N \leq M(t)$. Hence $\{N :\ g(N) \leq \gamma - 1\} = \{1, \ldots, M(\gamma - 1)\}$, so the least N with $g(N) \geq \gamma$ is $M(\gamma - 1) + 1$; as γ lies in the range of g, that least N satisfies $g(N) = \gamma$. □

Lemma 4.6 is used in Section 6 to recognize when a minimizer sits at the start of its plateau.

5 The shortest side and rigidity at exceptional plateau starts

Throughout this section S is an edge-perimeter minimizer with $n = |S|$ and $\sigma = \sigma(S) = g(n)$; by Theorem 4.4(d) it is directionally convex, so its support traces r_i and $\rho(S) = \min_i r_i$ are defined, and by Lemma 3.4(iii) $\rho(S) \geq 1$. Let $H = H(S)$.

A word on terminology, since four related phrases occur side by side from here on. The six *extreme lines* Λ_i of S (Section 2.3) are the same six lines as the *side lines* of H (Lemma 3.4(ii)). On each of them there are two sets: the *side* $H \cap \Lambda_i$ of the hull, of length ℓ_i, and the *support trace* $S \cap \Lambda_i$ of the configuration, of length r_i. A *shortest side* refers to $\min_i \ell_i = h$, a *shortest extreme line* or *shortest support trace* to $\min_i r_i = \rho(S)$. The theorem of Section 6 deletes a shortest support trace; the rigidity argument below compares it with the sides.

Observation 5.1. Let S be a directionally convex polyhex with tight hull H.

(a) For every i, $S \cap \Lambda_i \subseteq H \cap \Lambda_i$, so $r_i \leq \ell_i$, and $r_i = \ell_i$ holds if and only if $S \cap \Lambda_i = H \cap \Lambda_i$, that is, the whole side lies in S.

(b) If the two sides adjacent to $H \cap \Lambda_i$ in the cyclic order, namely $H \cap \Lambda_{i-1}$ and $H \cap \Lambda_{i+1}$, are contained in S, then so is the side $H \cap \Lambda_i$.

Proof. (a) is Lemma 3.4(i) restricted to Λ_i: two finite sets, one inside the other, have the same cardinality exactly when they coincide. For (b), the corners $\kappa_{i-1} = H \cap \Lambda_{i-1} \cap \Lambda_i$ and $\kappa_i = H \cap \Lambda_i \cap \Lambda_{i+1}$ lie on the two complete sides and hence in S. By Lemma 3.5(i) they are the two end cells of the run $H \cap \Lambda_i$; the trace $S \cap \Lambda_i$ is a run containing both ends of that run, so it contains the whole of it. □

Lemma 5.2. $\rho(S) \leq \left\lfloor \dfrac{\sigma + 3}{6} \right\rfloor$. *In words: the shortest support trace has at most one sixth of $\sigma + 3$ cells.*

Proof. By Lemma 3.4(ii) and (4), $\sum_i \ell_i = \lambda(H)+3 = \sigma+3$. Since $S \subseteq H$, $r_i \le \ell_i$, so $\sum_i r_i \le \sigma+3$; the minimum of six integers is at most their mean. □

The bound is attained by hulls whose six sides are as equal as they can be, and this is exactly what happens at the start of certain plateaux. The next proposition, the only delicate step of the paper, says that at those plateau starts a minimizer cannot have all six support traces that long.

Proposition 5.3 (rigidity at exceptional plateau starts)**.** *Let $m \ge 1$, let γ be in the range of g with $\gamma \in \{6m+3,\ 6m+5\}$, and let S be an edge-perimeter minimizer with $|S| = N_b(\gamma)$. Then $\rho(S) \le m$.*

In words: at the start of a plateau whose threshold is 3 or 5 modulo 6, the shortest support trace of a minimizer is strictly shorter than the bound of Lemma 5.2, which there equals $m+1$.

Proof. Put $N = N_b(\gamma)$. By Lemma 4.6, $N = M(\gamma-1)+1$, which for the two residues gives

$$N_b(6h-3) = 3h^2-4h+2, \qquad N_b(6h-1) = 3h^2-2h+1 \qquad (h = m+1), \tag{8}$$

since $\big((6h-4)^2+3\big)/12$ has integer part $3h^2-4h+1$ and $\big((6h-2)^2+3\big)/12$ has integer part $3h^2-2h$. By Theorem 4.4(d), S is directionally convex with $\lambda(S) = \gamma$, hence $\lambda(H) = \gamma$ by Lemma 3.4(ii).

Case $\gamma \equiv 3 \pmod 6$. Write $\gamma = 6h-3$, so $h = m+1 \ge 2$. We must show $\rho(S) \le h-1$. Suppose $\rho(S) \ge h$. Then every support trace has at least h cells, so every side of the hull has $\ell_i \ge r_i \ge h$; as $\sum_i \ell_i = \gamma+3 = 6h$ by (4) and Lemma 3.4(ii), all six sides equal h. Solving $\ell_2 = v+1 = \ell_4 = u+1 = \ell_6 = w+1 = h$ and $\ell_1 = L-v-w = h$ gives

$$u = v = w = h-1, \qquad L = 3h-2,$$

so by Lemma 3.5(iv)

$$|H| = T(3h-2) - 3\,T(h-1) = 3h^2-3h+1.$$

By (8), $N = 3h^2-4h+2$, so $|H|-N = h-1 \ge 1$. But $r_i \ge h = \ell_i$ for every i means, by Observation 5.1(a), that every side of the hull lies entirely in S, and Lemma 3.7 then gives $S = H$, contradicting $|S| = N < |H|$.

Case $\gamma \equiv 5 \pmod 6$. Write $\gamma = 6h-1$, so again $h = m+1 \ge 2$, and suppose $\rho(S) \ge h$. As before every side has $\ell_i \ge h$, while now $\sum_i \ell_i = \gamma+3 = 6h+2$; hence the shortest side has length exactly h and $\sum_i x_i = 2$. By Lemma 3.6 exactly one pair of opposite sides has length $h+1$. The three families are permuted cyclically by the 120° rotation of the lattice, which maps polyhexes to polyhexes, preserves adjacency, perimeter and the six extreme lines as a set, and maps the tight hull of S to the tight hull of the rotated S (the normalization of Definition 3.2 is re-applied after rotating); so we may assume that pair is (ℓ_1, ℓ_4), the two sides of the q-family. Solving $\ell_4 = u+1 = h+1$, $\ell_2 = v+1 = h$, $\ell_6 = w+1 = h$ and $\ell_1 = L-v-w = h+1$ gives

$$u = h, \qquad v = w = h-1, \qquad L = 3h-1$$

(one checks $\ell_3 = \ell_5 = h$ and $\lambda(H) = 3L-(u+v+w) = 6h-1 = \gamma$), and

$$|H| = T(3h-1) - T(h) - 2\,T(h-1) = 3h^2-h.$$

By (8), $N = 3h^2-2h+1$, so again $|H|-N = h-1 \ge 1$.

From $r_i \ge h$ and Observation 5.1(a) it follows that the four sides of length h, namely $H \cap \Lambda_i$ for $i = 2,3,5,6$, are contained in S. The long side $H \cap \Lambda_1$ is adjacent in the cyclic order to $H \cap \Lambda_6$ and $H \cap \Lambda_2$, both contained in S, so Observation 5.1(b) puts $H \cap \Lambda_1$ in S as well; the opposite side $H \cap \Lambda_4$ is handled in the same way through its neighbours $H \cap \Lambda_3$ and $H \cap \Lambda_5$. So all six sides are contained in S, and Lemma 3.7 gives $S = H$, contradicting $|S| = N < |H|$. □

The logical role of Proposition 5.3 in the next section is a single integer. Counting alone (Lemma 5.2 together with $n \geq M(\sigma-1)+1$) gives, after deleting a shortest support trace of length ρ, the bound $n - \rho \geq M(\sigma-2)+1$ in four residue classes of σ modulo 6, but only the weaker $n-\rho \geq M(\sigma-2)$ when $\sigma \equiv 3, 5 \pmod 6$, with equality exactly at a plateau start with $\rho = m+1$. The proposition rules out that equality case, and thereby raises the weak bound to the strict integer bound $n - \rho \geq M(\sigma-2)+1$ in those two classes as well. That one unit is what makes the peeled configuration a minimizer in Step 4 of the proof of Theorem 6.1.

6 The shortest-side skeleton theorem

Theorem 6.1 (shortest-side skeleton; Theorem 1.1)**.** *Let S be an edge-perimeter minimizer with $n = |S| \geq 3$, let $\sigma = \sigma(S) = g(n)$, and let Λ be* any *extreme line of S with $|S \cap \Lambda| = \rho(S)$. Put $S' := S \setminus \Lambda$. Then*

(i) S' is nonempty and is an edge-perimeter minimizer with $g(|S'|) = \sigma - 1$;

(ii) the cells of $S \cap \Lambda$ can be ordered $c_1, \ldots, c_\rho$ so that each $T_j := S' \cup \{c_1, \ldots, c_j\}$ is a directionally convex polyhex with $\mathrm{exc}(T_j) \leq 2$.

In words: peeling a shortest extreme line from a minimizer lands on a minimizer one threshold step down, and putting the line back one cell at a time never costs more than two units of edge perimeter.

Proof. By Theorem 4.4(d), S is directionally convex with $\lambda(S) = \sigma$, and $\rho = \rho(S) \geq 1$. Since $n \geq 3$, $g(3) = \lceil \sqrt{33} \rceil = 6$, and g is nondecreasing, we have $\sigma \geq 6$. Write $\sigma = 6m + s$ with $0 \leq s \leq 5$; then $m \geq 1$.

Step 1 (the two bounds). Lemma 5.2 gives $\rho \leq \lfloor (\sigma+3)/6 \rfloor$, and $g(n) = \sigma$ means $n > M(\sigma-1)$, that is

$$n \;\geq\; M(\sigma-1)+1. \tag{9}$$

Step 2 (the residue table). For $t = 6m + j$ with $0 \leq j \leq 5$, $M(t) = 3m^2 + mj + \lfloor (j^2+3)/12 \rfloor$, and $\lfloor (j^2+3)/12 \rfloor = 0, 0, 0, 1, 1, 2$ for $j = 0, \ldots, 5$. Taking differences (for $s = 0$ both arguments $\sigma - 1, \sigma - 2$, and for $s = 1$ the argument $\sigma - 2$, fall into the previous block $6(m-1)+j$), and using $\lfloor (\sigma+3)/6 \rfloor = m + \lfloor (s+3)/6 \rfloor$:

$s = \sigma \bmod 6$	$M(\sigma-1) - M(\sigma-2)$	$\lfloor (\sigma+3)/6 \rfloor$	difference
0	m	m	0
1	m	m	0
2	m	m	0
3	m	$m+1$	-1
4	$m+1$	$m+1$	0
5	m	$m+1$	-1

Combining the two bounds of Step 1, $n - \rho \geq M(\sigma-1) + 1 - \lfloor (\sigma+3)/6 \rfloor$. If $s \notin \{3, 5\}$ the table gives $M(\sigma-1) - \lfloor (\sigma+3)/6 \rfloor \geq M(\sigma-2)$, hence

$$n - \rho \;\geq\; M(\sigma-2)+1. \tag{10}$$

If $s \in \{3, 5\}$ the table gives only $n - \rho \geq M(\sigma-2)$, with equality if and only if (9) and Lemma 5.2 are both equalities: $n = M(\sigma-1)+1$ and $\rho = m+1$. In words: the strict inequality (10) already holds

except possibly when all three of the following occur at once — the threshold is 3 or 5 modulo 6, the configuration is at the very start of its plateau, and the shortest trace is as long as the counting allows.

Step 3 (the exceptional case is excluded by rigidity). Suppose $s \in \{3, 5\}$, $n = M(\sigma - 1) + 1$ and $\rho = m + 1$. By Lemma 4.6, $n = N_b(\sigma)$; also $\sigma = g(n)$ lies in the range of g and $m \geq 1$. So Proposition 5.3 applies with $\gamma = \sigma$ and gives $\rho \leq m$, contradicting $\rho = m + 1$. Hence (10) holds in every residue.

Step 4 (conclusion (i)). From (10) and the definition of g, $g(n - \rho) \geq \sigma - 1$. As $n - \rho \geq M(\sigma - 2) + 1 \geq 1$, the set S' is nonempty, so Lemma 4.1(i) applies: S' is a directionally convex polyhex with $\sigma(S') \leq \sigma - 1$. By Lemma 2.6 and Theorem 4.4(c),

$$2(\sigma - 1) \;\leq\; 2g(|S'|) \;=\; p_{\min}(|S'|) \;\leq\; p(S') \;=\; 2\sigma(S') \;\leq\; 2(\sigma - 1),$$

so all are equalities: S' is a minimizer and $g(|S'|) = \sigma - 1$. Nothing so far depended on which shortest line Λ was chosen.

Step 5 (conclusion (ii)). By Lemma 4.1(ii) there is an ordering $c_1, \ldots, c_\rho$ of $S \cap \Lambda$ such that every T_j is a directionally convex polyhex with $\sigma(T_j) \leq \sigma$. As $|T_j| \geq |S'|$ and g is nondecreasing, $g(|T_j|) \geq \sigma - 1$. Therefore

$$\operatorname{exc}(T_j) = p(T_j) - 2g(|T_j|) = 2\big(\sigma(T_j) - g(|T_j|)\big) \leq 2\big(\sigma - (\sigma - 1)\big) = 2.$$

In words: during the restoration the perimeter never exceeds the starting value while the threshold is already one step lower, so the excess is at most two. □

Corollary 6.2 (skeleton chain)**.** *Every edge-perimeter minimizer S admits a chain $S = T_0 \supset T_1 \supset \cdots \supset T_k$ of edge-perimeter minimizers with $|T_k| \leq 2$, each obtained from the previous one by deleting a shortest extreme line, and $g(|T_{i+1}|) = g(|T_i|) - 1$ for every step.*

Proof. Apply Theorem 6.1(i) while $|T_i| \geq 3$; sizes strictly decrease. □

Proof of Theorem 1.2. Take the chain of Corollary 6.2. If $|T_k| = 2$, then T_k is a domino, which has exc = 0, and either of its cells is a single cell with exc = 0; extend the chain by one step. Between consecutive members insert the configurations T_j of Theorem 6.1(ii), each a polyhex of excess at most 2. Reading the resulting sequence from the single cell up to S gives (b). For (a), each $S_{k+1} = S_k \cup \{c\}$ is connected and S_k is nonempty, so c has a neighbour in S_k; otherwise $\{c\}$ would be a connected component of S_{k+1} on its own. □

Remark 6.3 (the procedure). The construction is short to state: repeatedly delete a shortest extreme line, and read the resulting sequence of minimizers backwards, restoring each deleted line one cell at a time from a cell adjacent to what remains. The minimizers themselves form the skeleton of the growth; the excess 2 is spent only inside the restoration of a single line.

7 Optimality of the constant two

Corollary 6.2 says that the skeleton of the growth — the configurations at which a whole extreme line has just been restored — can always be taken to consist of minimizers. This section shows that the same cannot be asked of the intermediate one-cell stages: the constant 2 of Theorem 1.2 cannot be lowered. The argument is self-contained and uses no computation: a single identity for the effect of deleting one cell, applied to one explicit 13-cell minimizer.

7.1 The deletion identity

For a finite set of cells S and $c \in S$ write $\deg_S(c)$ for the number of cells of S adjacent to c.

Lemma 7.1. *For every finite set of cells S and every $c \in S$,*

$$p(S \setminus \{c\}) = p(S) - 6 + 2\deg_S(c). \tag{11}$$

In words: removing one cell takes away the six edges of that cell from the boundary, and gives back, for each neighbour the cell had inside S, the two sides of the shared edge that now lie on the boundary.

Proof. Write $n = |S|$ and $a = a(S)$. Deleting c removes one cell and exactly $\deg_S(c)$ adjacent pairs, so $|S \setminus \{c\}| = n - 1$ and $a(S \setminus \{c\}) = a - \deg_S(c)$. By the definition $p(K) = 6|K| - 2a(K)$,

$$p(S \setminus \{c\}) = 6(n-1) - 2\big(a - \deg_S(c)\big) = \big(6n - 2a\big) - 6 + 2\deg_S(c),$$

which is (11). □

No connectedness is assumed in Lemma 7.1; (11) is an identity for arbitrary finite sets of cells. Connectedness enters only when $S \setminus \{c\}$ is to be used as the predecessor of S in a growth chain, since a chain consists of polyhexes.

7.2 An explicit 13-cell minimizer

Example 7.2. Let

$$S^* := \{(0,r) : 1 \le r \le 4\} \ \cup\ \{(1,r) : 0 \le r \le 4\} \ \cup\ \{(2,r) : 0 \le r \le 3\}.$$

In words: three consecutive q-lines carrying 4, 5 and 4 cells respectively, each a run, arranged so that the whole set is a polyhex.

Lemma 7.3. *The set S^* is a directionally convex polyhex with*

$$|S^*| = 13, \qquad p(S^*) = 26 = p_{\min}(13),$$

so S^* is a 13-cell minimizer. Its degree distribution is

$$3^6\, 4^4\, 6^3,$$

in words: six cells of degree 3, four of degree 4 and three of degree 6; in particular S^* has no cell of degree 2 or less.

Proof. The three traces in the q-family are the runs $r \in [1,4]$, $r \in [0,4]$ and $r \in [0,3]$, giving $|S^*| = 4 + 5 + 4 = 13$. The traces in the r-family are

$$r = 0 :\ q \in [1,2], \quad r = 1,2,3 :\ q \in [0,2], \quad r = 4 :\ q \in [0,1],$$

and those in the s-family, ordered by q, are

$$s = 1 :\ q \in [0,1], \quad s = 2,3,4 :\ q \in [0,2], \quad s = 5 :\ q \in [1,2].$$

Every trace is a single run, so S^* is directionally convex, and it is connected because consecutive q-traces overlap in their r-ranges. Counting the occupied lines gives $\lambda(S^*) = 3 + 5 + 5 = 13$, so by Lemma 2.6 $p(S^*) = 2\lambda(S^*) = 26$. Since $12 \cdot 13 - 3 = 153$ and $12^2 < 153 \le 13^2$, we have $g(13) = 13$ and $p_{\min}(13) = 26$.

For the degrees, the three cells $(1,1), (1,2), (1,3)$ have all six neighbours in S^*; the four cells $(0,2), (0,3), (2,1), (2,2)$ have exactly four; and the six cells $(0,1), (0,4), (1,0), (1,4), (2,0), (2,3)$ have exactly three. These thirteen cells exhaust S^*, and $6 \cdot 3 + 4 \cdot 4 + 3 \cdot 6 = 52 = 2a(S^*)$, consistent with $p(S^*) = 6 \cdot 13 - 52 = 26$. □

7.3 No 12-cell minimizer predecessor

Proposition 7.4. *For every $c \in S^*$,*

$$p(S^* \setminus \{c\}) \;\geq\; 26 \;=\; p_{\min}(12) + 2.$$

In particular no one-cell deletion from S^* is a 12-cell minimizer.

Proof. Since $12 \cdot 12 - 3 = 141$ and $11^2 < 141 \leq 12^2$, we have $g(12) = 12$ and $p_{\min}(12) = 24$. By Lemma 7.1 and $p(S^*) = 26$,

$$p(S^* \setminus \{c\}) = 26 - 6 + 2\deg_{S^*}(c) = 20 + 2\deg_{S^*}(c),$$

in words: the perimeter after the deletion is twenty plus twice the degree of the deleted cell. Equality $p(S^* \setminus \{c\}) = 24$ would force $\deg_{S^*}(c) = 2$, and by Lemma 7.3 no cell of S^* has degree 2. Every degree is at least 3, so $p(S^* \setminus \{c\}) \geq 26$ for every c. □

Note that connectedness of $S^* \setminus \{c\}$ never had to be examined: if the deletion disconnects S^*, the result is not a polyhex and cannot be a predecessor in a growth chain; if it does not, Proposition 7.4 already rules it out as a minimizer. Thirteen separate case checks are not needed.

7.4 The constant 2 is optimal

Proof of Theorem 1.3. Suppose there were a chain of polyhexes $S_1 \subset \cdots \subset S_{13} = S^*$ with $|S_k| = k$ and $\mathrm{exc}(S_k) = 0$ for every k. Then S_{12} is a polyhex with twelve cells obtained from S^* by deleting a single cell, and $\mathrm{exc}(S_{12}) = 0$ means $p(S_{12}) = p_{\min}(12) = 24$. This contradicts Proposition 7.4.

It remains to see that no uniform bound strictly between 0 and 2 is available. By Lemma 2.6 the edge perimeter of any finite set of cells is even, and $p_{\min}$ is even by Theorem 4.4(c); hence

$$\mathrm{exc}(K) = p(K) - p_{\min}(|K|) \in \{0, 2, 4, \dots\}$$

for every polyhex K, in words: the excess is always a nonnegative even integer. A uniform bound of 1 is therefore the same requirement as a uniform bound of 0, which has just been refuted. So 2 is the least admissible constant. □

Remark 7.5 (a second proof of Proposition 7.4). Proposition 7.4 also follows from line counting, which explains the example from a different side. As computed in the proof of Lemma 7.3, every one of the thirteen lines occupied by S^* carries at least two cells: the shortest traces are $r = 0$, $r = 4$, $s = 1$ and $s = 5$, each with exactly two. Deleting a single cell therefore leaves every occupied line still occupied, so $\lambda(S^* \setminus \{c\}) = \lambda(S^*) = 13$, and Lemma 2.6 gives

$$p(S^* \setminus \{c\}) \;\geq\; 2\lambda(S^* \setminus \{c\}) \;=\; 26.$$

In words: no single deletion can vacate a line, so the perimeter cannot drop below twice the unchanged line count. The degree argument of Section 7.3 is shorter and is the one we take as canonical; this one shows that the obstruction can be read off from the support structure rather than from local degrees.

Computational enumeration shows that $N = 13$ is the smallest size at which this obstruction occurs; see Section 9.

8 Remarks

Remark 8.1 (excess alone is not a usable state space). Theorem 1.1 runs its induction entirely inside the class of minimizers, deleting a whole extreme line at a time. One might ask whether a simpler induction, deleting one cell at a time inside the class $\{\mathrm{exc} \le 2\}$, could work instead. It cannot. Let H be the regular hexagon $H(10; 3, 3, 3)$, which has 37 cells and $\sigma(H) = 21$, and let X_{31} be H with its six corner cells removed. Each corner cell is an end cell of the trace of H on each of the three lines through it (it is a vertex of the convex region whose cells make up H), so removing the corners shortens traces only at their ends. Thus X_{31} is a directionally convex polyhex with $|X_{31}| = 31$, $\sigma(X_{31}) = 21$, $g(31) = 20$ and $\mathrm{exc}(X_{31}) = 2$; yet every line meeting X_{31} carries at least two of its cells, so deleting any single cell leaves all 21 lines occupied and, since $g(30) = 19$, an excess of at least $2(21 - 19) = 4$. Within $\{\mathrm{exc} \le 2\}$ there is no one-cell move down from X_{31}. The skeleton theorem avoids this by never leaving the minimizers except during the restoration of a single line, where the excess is controlled by Theorem 6.1(ii) rather than by a local search.

Remark 8.2 (a wider class). The peeling argument of Section 6 can be run on a larger class than the minimizers, namely the directionally convex polyhexes S with $|S| \ge M(\sigma(S) - 1)$, and yields one-cell growth with excess at most two for all of them. The wider class is also closed under shortest-line peeling, but proving that closure requires a stronger quantitative peeling estimate, obtained from a finite case analysis of the hull, rather than the plateau-start rigidity of Proposition 5.3; the proof is correspondingly longer. Since the main theorem does not need the wider class, that argument is recorded in a separate note.

Remark 8.3 (a one-cell-at-a-time proof). Theorem 1.2 also admits a proof that descends one cell at a time, keeping the configuration directionally convex with at most one occupied line above the threshold and tracking two competing budgets, the length of the shortest extreme line and the distance to the next drop of the threshold. That proof uses the same rigidity at exceptional plateau starts as Proposition 5.3, and in addition a local lemma showing that a degree-two articulation cell forces excess at least four. It is recorded separately as well.

9 Computational verification

No theorem of this paper depends on finite enumeration. Computation is used here for two purposes only: to determine one size that the analytic arguments do not determine (Section 9.1), and to re-check the statements of the paper with code written independently of their proofs (Section 9.2).

Two different senses of "independent" should be kept apart. The *implementations* are independent of the proofs: the hull identities are re-derived symbolically and by enumeration, the peeling lemma is re-checked cell by cell, and so on, without following the steps of the argument. The *enumerations of minimizers*, on the other hand, are not logically independent of the analysis: they generate the candidates through Theorem 4.4(d) (a minimizer is directionally convex with $\lambda = g(N)$) and Section 9.3 says exactly how. A row that checks a statement on "all minimizers with $N \le 34$" is therefore a check conditional on that theorem, and two implementations of such a check are two implementations, not two proofs. Checks that run over all polyhexes up to a given size carry no such condition.

Each entry below carries one of three labels.

ANALYTIC	The statement is proved analytically in the body; the computation is a re-check and the proof does not rest on it.
EXACT	The computation enumerates a finite set completely, so it is conclusive *for that finite set*. No general theorem is made to depend on it.
CHECK	A consistency check up to a finite size, or a census. It is not evidence for any general statement.

9.1 A size the proofs do not determine

Section 7 exhibits one 13-cell minimizer with no all-minimizer growth chain and proves analytically that the constant 2 cannot be lowered. It says nothing about whether smaller sizes behave the same way. That is settled by enumeration.

For each $N \leq 13$ we enumerated $\mathcal{M}_N$ up to translation and counted the configurations reachable from a single cell by a growth all of whose intermediate stages are minimizers.

N	$\lvert\mathcal{M}_N\rvert$	reachable	unreachable	N	$\lvert\mathcal{M}_N\rvert$	reachable	unreachable
1	1	1	0	8	6	6	0
2	3	3	0	9	27	27	0
3	2	2	0	10	3	3	0
4	3	3	0	11	24	24	0
5	6	6	0	12	2	2	0
6	14	14	0	13	27	18	**9**
7	1	1	0				

So $N = 13$ is the smallest size at which an unreachable minimizer occurs, and the configuration S^* of Section 7 is one of the nine. The 27 configurations of size 13 form 4 equivalence classes under the point group of the lattice, and the 18 reachable ones form 2 such classes. The enumeration is complete for every $N \leq 13$, so the statement that 13 is the smallest such size is settled outright for that finite range. That minimality statement is a computational fact and is not part of Theorem 1.3, which needs one 13-cell example and is proved analytically. EXACT

At $N = 34$, of the 392 translation-normalized minimizers (39 free equivalence classes), 102 (9 free classes) are reachable by an all-minimizer growth. CHECK

9.2 Consistency checks

Each row was carried out with an implementation written independently of the proof it re-checks. Row 7 and the census of Section 9.1 generate their minimizers through Theorem 4.4(d) and are conditional on it in the sense explained at the start of this section; every other row enumerates all polyhexes, all directionally convex polyhexes or all admissible hexagons in its range, or examines a named configuration exhaustively, and carries no such condition. No violation was found in any of them.

	what is checked	range	status
1	$p = 2R \geq 2\lambda$, and equality exactly for directionally convex sets	all polyhexes with $N \leq 11$	ANALYTIC
2	the hull description and the defect identity of Section 3	symbolically in L, u, v, w; all 5,220 admissible hexagons with $L \leq 16$	ANALYTIC
3	$\lvert S\rvert \leq M(\sigma(S))$; minimizers are directionally convex with $\lambda = g$	all polyhexes with $N \leq 11$	ANALYTIC
4	$p_{\min}(N) = 2\lceil\sqrt{12N-3}\rceil$	brute force for $N \leq 9$; against the capacity function for $N \leq 3000$	ANALYTIC
5	peeling: connectedness, directional convexity and the drop in σ	all 145,374 side peelings of directionally convex polyhexes with $N \leq 10$	ANALYTIC
6	the residue table of Section 6 and the two plateau-start counts (8)	$\sigma \leq 2999$; $m \leq 59$	ANALYTIC
7	Theorem 6.1: every shortest-line deletion from a minimizer lands on a minimizer with g down by one, and the restoration order keeps exc ≤ 2	all 2,382 shortest-line deletions from minimizers with $3 \leq N \leq 34$, ties included; two independently written implementations, both enumerating minimizers via Theorem 4.4(d)	ANALYTIC
8	the degree table of S^* and its thirteen one-cell deletions	exhaustive for S^*	EXACT
9	the configuration X_{31} of Remark 8.1	exhaustive	EXACT

For orientation: among the 6,498 deletions of a *non*-shortest extreme line from the same minimizers, only 3,192 land on a minimizer. The word "shortest" in Theorem 6.1 is not decorative.

9.3 Reproduction

The enumeration of $\mathcal{M}_N$ uses Theorem 4.4(d): a minimizer is directionally convex with $\lambda = g(N)$, so it suffices to enumerate the tight hulls with the corresponding edge semiperimeter and then the subsets that keep it. For $N \leq 34$ one has $\lvert H\rvert - N \leq 3$. Reachability is then decided by dynamic programming on the size. The census of Section 9.1 is thus exact for the class of configurations that Theorem 4.4(d) describes; it does not re-establish that theorem.

The scripts reproducing every number quoted above are provided as ancillary files; they use the Python standard library only and need no input files.

Statement on the use of AI-assisted tools

AI-assisted tools (Claude and ChatGPT) were used during proof search, counterexample search, computational verification, auditing, and preparation of the manuscript. The author evaluated the resulting arguments and computations and is responsible for the mathematical content of the paper.

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